

Day 1 — Sat 5 Sep

Countdown: 21 days to General English & GK

Method: start by recalling yesterday's topic with this page closed. Study the sections below, work every example by hand, then do 20 MCQs.

● General — playbook

Source: [General/01_GENERAL_PAPER_PLAYBOOK.md](#)

Paper 1 — General English & General Knowledge: The Playbook

Exam: NPSC Combined Technical Services Examination (CTSE) 2026

Paper: General English & General Knowledge (Combined) — **100 marks, 3 hours**

Date: 26 September 2026, 09:00–12:00

Written for: a B.Tech CS candidate with a strong technical base and limited humanities practice.

This is the **one-shot master guide**. If you only read one file, read this one.

The other two files ([02_ESSAY_BANK.md](#), [03_GK_CAPSULE.md](#)) are the raw material this playbook tells you how to use.

0. What the paper actually is (read this first)

Straight from the official pattern ([Materials/_Shared/OFFICIAL_SYLLABUS.md](#), §1):

#	Component	Marks	It tests
2.1.1	Essay (1 essay)	20	Sustained written argument in English
2.1.2	Précis writing (1 passage)	10	Comprehension + compression
2.1.3	Amplification (1 maxim/proverb)	10	Interpretation + illustration
2.1.4	Construction of sentence & Grammar	10	Mechanical accuracy
2.2	General Knowledge	50	Current events, general science, mental ability, Indian history, Indian polity, Indian economy, Geography — <i>"combination of objective type and short descriptive answers"</i>
	Total	100	in 180 minutes

Three consequences you must internalise

1. **Half the paper is GK, and GK is the only half where marks are binary.**

In English, a mediocre essay still gets ~9–11/20. In GK, you either know the answer or you do not. So GK is where your 21 days of *study* go; English is where your 21 days of *practice* go. They are different activities. Do both.

2. **English is scored generously in the middle and stingily at the top.**

Almost nobody scores 17/20 on the essay. Almost everybody who writes 900 clean,

relevant, well-structured words scores 12–14. Your target is **not** a brilliant essay. Your target is a *competent, complete, legible* essay — and then to bank every mechanical mark in précis, amplification and grammar. That is a very achievable 35–40 out of 50 in English.

3. "**Combination of objective type and short descriptive answers**" means GK is probably split — a block of one-mark/two-mark objective items plus several 3–5 mark short-answer questions. Prepare for both. Objective = recall of facts. Short descriptive = a 60–120 word structured answer. Section 6 of this file teaches the short-descriptive format.

1. Exam-day time allocation — the 180-minute plan

1.1 The recommended order of attempt

Do not attempt in the order printed on the paper. Attempt in this order:

Order	Component	Marks	Minutes	Cumulative
1st	Grammar / sentence construction	10	15	15
2nd	GK — objective portion	~20–25	25	40
3rd	GK — short descriptive portion	~25–30	35	75
4th	Amplification	10	20	95
5th	Précis	10	25	120
6th	Essay	20	50	170
—	Buffer / review / page numbering	—	10	180

1.2 Why this order

- **Grammar first (15 min).** These are the cheapest marks on the paper — pure recall, no thinking, no fatigue cost. Banking 8–9/10 in the first fifteen minutes settles your nerves and guarantees a floor. Never leave grammar until the end; if you run out of time you will lose marks you already knew.
- **GK second, while your memory is freshest.** Recall degrades under fatigue far faster than writing fluency does. Do the objective block fast — if you do not know an item in 20 seconds, mark it, move on, come back in the buffer.
- **Amplification and précis in the middle.** Both are *bounded* tasks — there is a correct-ish answer and a fixed length. They do not expand to fill available time the way an essay does. Doing them before the essay stops the essay from eating them.
- **Essay last, with a hard 50-minute block.** The essay is the one task where you can keep writing forever. Putting it last means the clock, not your judgement, ends it — and a 50-minute essay written at the end still scores 12–14 if you follow the skeleton. A rushed 15-minute précis at the end scores 4.

One exception: if the essay topic list is on page 1 and you can see it, *read the essay topics in the first 60 seconds* and pick your topic. Then let it marinate in the back of your mind while you do everything else. You will start the essay with your argument already half-formed. This single trick is worth 2–3 marks.

1.3 Micro-timings inside each block

Grammar (15 min) — 45–60 seconds per item. Do not agonise. If two options both

"sound right", pick the shorter/simpler one and move on.

GK objective (25 min) — Two passes. Pass 1 (18 min): answer everything you know instantly. Pass 2 (7 min): return to the marked ones and make an educated guess.

There is no stated negative marking on this paper (negative marking is specified only for the Technical MCQ paper) — so **attempt every question**. Leave nothing blank.

GK short descriptive (35 min) — Budget by marks: roughly 5 minutes per 5-mark question, 3 minutes per 3-mark. Write to length, then stop.

Amplification (20 min) — 3 min planning, 15 min writing, 2 min checking.

Précis (25 min) — 6 min reading and marking the passage, 4 min drafting in the margin, 12 min writing the fair copy, 3 min counting words and writing the title.

Essay (50 min) — **8 min planning on rough paper**, 38 min writing, 4 min reading back. The 8 minutes of planning is non-negotiable. An unplanned essay loses more than 8 minutes' worth of marks in repetition and drift.

1.4 The three clock checkpoints

Write these on your rough sheet the moment you get it:

```
10:00 → Grammar done, GK objective underway
11:15 → ALL GK done. Nothing else touched.
11:40 → Amplification + précis done. Essay starts NOW.
12:00 → Pens down.
```

If at 11:40 you are not on the essay, **abandon whatever you are doing and start the essay**. Twenty marks unattempted is catastrophic; a half-finished précis is not.

2. ESSAY — 20 marks

The single biggest block on the paper. Fifty minutes, one-fifth of your total.

2.1 How state PSC essays are actually marked

Examiners rarely have a public rubric, but the working marking scheme across state commissions is consistently something like this. Treat it as the shape of the thing, not gospel:

Dimension	Weight	What the examiner is looking for
Content & relevance	~7/20	Do you address <i>this</i> topic? Is there real substance — facts, examples, dimensions — or padding?
Structure & coherence	~5/20	Intro → body with distinct paragraphs → conclusion. One idea per paragraph. Logical flow.
Language & expression	~4/20	Grammar, vocabulary, sentence variety, no repetition.
Balance & maturity	~2/20	Do you see more than one side? Is the tone measured rather than ranting?
Presentation	~2/20	Legibility, margins, paragraph breaks, no overwriting, a visible plan.

The two commonest reasons a competent candidate scores 9 instead of 14:

- They wrote about the *general area* of the topic instead of the *exact* topic. ("Role of technology in governance" answered as an essay on "what is technology".)
- They wrote one enormous 800-word block with no paragraph breaks.

Both are free to fix.

2.2 Length

Target word count	800–1000 words
Absolute floor	600 words (below this it reads as unfinished)
Ceiling	1200 words (beyond this you are stealing time from other sections)
Rough page count	3.5–4.5 sides of standard answer-book paper in normal handwriting
Paragraphs	6–9

Calibrate this **before the exam**: write one timed 900-word essay at home, count the lines, and memorise "900 words = N lines in my handwriting". On exam day you then never count words — you count lines. Saves five minutes.

2.3 The reusable skeletons

You will walk into the hall with **two skeletons memorised**. Whatever the topic, one of them fits.

Skeleton A — the 5-paragraph (use when the topic is narrow or you are short on ideas)

¶	Function	Words
1	Introduction — hook, definition/context, your thesis, roadmap	120
2	The case / the positives / the current situation	180
3	The counter-case / problems / challenges	180
4	The way forward — concrete, specific suggestions	180
5	Conclusion — synthesis + forward-looking close	110
	Total	~770

Skeleton B — the 7-paragraph (use when the topic is broad; this is your default)

¶	Function	Words
1	Introduction — hook / definition / thesis / roadmap	130
2	Background & context — how did we get here? (the <i>past</i>)	130
3	Dimension 1 — usually Economic	140
4	Dimension 2 — usually Social / Human	140
5	Dimension 3 — usually Political/Governance, Technological, or Environmental	140
6	Challenges & the balanced view — "however..."	140
7	Way forward + Conclusion (can be split into two short paragraphs)	150
	Total	~970

Add a Nagaland / North-East paragraph wherever it plausibly fits. On almost any NPSC essay topic — digital governance, education, unemployment, environment, women, tourism, infrastructure — you can insert one paragraph localising the argument to Nagaland or the North-East. Examiners in a state commission notice this and reward it. It is probably the highest-return single habit in this whole document.

2.4 Writing an introduction that scores

A scoring introduction does **four** things in about 120 words, in this order:

1. **Hook** — one arresting sentence. Options:
 - A short relevant quotation.
 - A stark contrast: *"India launches missions to the Moon while a third of its children remain stunted."*
 - A definition: *"Digital governance is the use of information technology to deliver public services more transparently and more cheaply."*
 - A scene: *"In a village in Nagaland's Kiphire district, a mobile signal arrives for two hours a day."* (Use only if plausible — do not invent a specific claim.)
2. **Context** — two sentences placing the topic in time and place.
3. **Thesis** — one sentence stating what *you* will argue. This is the sentence the examiner looks for. *"This essay argues that X is not primarily a technological problem but an institutional one."*
4. **Roadmap** — one sentence: *"It examines the economic, social and governance dimensions in turn, before turning to the specific case of the North-East."*

Never open with "Since the dawn of civilisation, man has..." or "In this modern era of globalisation...". These are the two most common openings on any answer script and examiners are numb to them.

2.5 Writing a conclusion that scores

Rules:

- **Never** introduce a new argument.
- **Never** just repeat the introduction in different words.
- Do three things in ~110 words:
 1. **Synthesise** — one sentence pulling the body together ("The record is therefore mixed: real gains in access, persistent gaps in quality.").
 2. **Position** — restate your thesis, now as something you have *earned*.
 3. **Forward look** — end optimistic but conditional. "...provided that...", "...if X is matched by Y...". A conditional optimistic close reads as mature; an unqualified "thus India will surely become a superpower" reads as naive.
- A short closing quotation is fine if it genuinely fits. One quotation per essay is plenty; three is showing off.

2.6 Generating content fast under pressure

This is the skill that separates a 9 from a 14 when you don't know much about the topic. Spend your 8 planning minutes running **two frameworks** over the topic and harvesting whatever falls out.

Framework 1 — PESTLE (gives you *dimensions* → your body paragraphs)

Letter	Lens	Ask yourself
P	Political / Governance	Who decides? Which ministry, law, scheme? Centre vs State?
E	Economic	Cost, jobs, growth, inequality, who pays, who profits
S	Social	Families, women, youth, tribes, caste, migration, health, education
T	Technological	What technology enables or threatens this? Digital divide?
L	Legal / Ethical	Which rights are engaged? Article 21? Privacy? Consent?

E

Environmental

Land, forests, rivers, climate, biodiversity, jhum cultivation

Write P-E-S-T-L-E vertically on your rough sheet, spend 4 minutes writing 1–2 bullets next to each, then **pick the best three or four** as your body paragraphs. You now have an essay. You have not needed a single memorised fact.

Framework 2 — Past / Present / Future (gives you *movement*)

Every good essay moves in time.

- **Past:** how did this come about? (§2 of Skeleton B)
- **Present:** what is the situation now, with its tensions? (body)
- **Future:** what should be done, and by whom? (way forward)

An essay with all three feels complete. An essay stuck in the present feels thin.

Framework 3 — the stakeholder sweep (use when PESTLE runs dry)

List who is affected: **citizen / farmer / student / woman / youth / worker / business / state government / central government / civil society / future generations.**

Each stakeholder is a potential paragraph or at minimum a sentence.

The 8-minute planning ritual (do exactly this)

```
min 0–1  Read all topic options. Pick. Underline the KEY NOUN and the KEY VERB
         in the topic. ("ROLE of technology in RURAL development" – you must
         discuss role, and you must discuss rural.)
min 1–5  PESTLE dump on rough sheet. Past/Present/Future dump.
min 5–7  Choose 5–7 paragraphs. Number them. Write ONE phrase per paragraph
         = your topic sentence.
min 7–8  Write your thesis sentence in full, and your closing line in full.
         (Knowing your last line before you start is a huge stabiliser.)
```

Then write. Do not deviate from the plan even if a better idea arrives at minute 30.

2.7 Using facts, data and quotes

- **A few precise facts beat many vague ones.** Two or three concrete anchors in a 900-word essay is enough. Ten unsupported statistics look invented.
- **Only use numbers you are sure of.** A wrong statistic is worse than no statistic — it signals you are bluffing. If you are unsure, use a *qualitative* claim instead: "a large share of", "the majority of", "roughly half". These cannot be wrong.
- **Safe anchors you can reliably use** (verify the ones you plan to lean on):
- Constitutional articles: Art. 14 equality, Art. 19 freedoms, Art. 21 life and personal liberty, Art. 21A education, Art. 51A fundamental duties, **Art. 371A special provisions for Nagaland**, Art. 243 Panchayati Raj.
- Named schemes and missions (no numbers needed): Digital India, MGNREGA, Ayushman Bharat, PM Jan Dhan Yojana, Swachh Bharat, National Education Policy 2020, Act East Policy, Skill India, Startup India, Jal Jeevan Mission.
- Institutions: NITI Aayog, RBI, Election Commission, CAG, Supreme Court, NEC (North Eastern Council), DoNER Ministry.
- Nagaland-specific: Article 371A; the Nagaland Village Council & Village Development Board system; the Communitisation of Public Institutions and Services Act, 2002 (a genuinely distinctive Nagaland innovation and an excellent essay example); Hornbill Festival; Act East Policy and the state's border with Myanmar.
- **Quotations:** one is enough, two maximum. See [02_ESSAY_BANK.md](#) for a bank of 20. Attribute correctly or attribute vaguely ("as an old proverb has it") — never misattribute to a famous name.

2.8 Common mistakes that lose marks

Mistake	Cost	Fix
No paragraph breaks / one giant block	−3	Break every 120–150 words
Ignoring half the topic ("causes and remedies")	−4	Underline every clause in the question
One-sided rant	−2	Always one "However..." paragraph
Repeating the same idea in ¶3 and ¶6	−2	Follow the plan; each ¶ has one job
Very long sentences (40+ words)	−2	Aim for average 15–20 words; vary length
Overwriting, scratching out, no margin	−2	Cross out with a single clean line
Running to 1500 words and losing time elsewhere	−6 elsewhere	Line-count discipline
Sweeping unverifiable statistics	−2	Qualitative language instead
Slang, SMS abbreviations, "&", "etc." repeatedly	−1	Formal register throughout
Starting with "Since time immemorial..."	−1	Use a real hook

2.9 Handwriting and presentation

Presentation is worth roughly 2 marks and costs nothing.

- **Legibility beats beauty.** If the examiner has to slow down, you lose.
- Leave a **left margin** and a blank line between paragraphs.
- **Underline your essay title.** Underline nothing else except, at most, 2–3 key terms. An essay with fifteen underlines looks frantic.
- Write the **word count at the end** if you are near target — it signals control. Skip it if you overshot.
- Use **black or blue ink only**; one colour throughout.
- If you must delete, one horizontal line. Never scribble a blob.
- **Numbering:** label your answers clearly with the question number. An unlabelled answer can be missed entirely.

2.10 Practice plan for the essay in 21 days

Days 1–3	Read the 4 model essays in 02_ESSAY_BANK.md . Copy out one introduction and one conclusion by hand to internalise the rhythm.
Days 4–14	One full timed essay every alternate day (5 total). 50 minutes, no phone, handwritten. Pick from the topic bank, alternating themes.
Days 4–14 (off days)	10-minute drill: pick a topic, do <i>only</i> the 8-minute plan. No writing. Do 8–10 of these. Plans are cheap and build the reflex.
Days 15–19	2 more timed essays; one must be a Nagaland/North-East topic.
Days 20–21	No new essays. Re-read your own plans and the quotation bank.

3. PRÉCIS WRITING — 10 marks

3.1 What a précis is, in plain English

You are given a passage of maybe 350–450 words. You must rewrite it in about **one-third** of its length, keeping *every essential idea* and *nothing else*, in **your own words**, in **your own continuous prose**, and give it a **title**.

It is not a summary you write casually. It has strict conventions, and most of the 10 marks are for **following the conventions**, not for elegance. This is the single most *mechanically learnable* item on the English half of the paper. A candidate who memorises the rules should score 7–8/10 reliably.

3.2 The strict rules

#	Rule	Note
1	Length $\approx$ one-third of the original.	If the passage is 360 words, write 115–125. Tolerance is roughly $\pm 10\%$. Being 50% too long is the commonest failure.
2	Always give a title.	Short — 3 to 7 words. Written above, centred or left, underlined. A missing title costs a mark outright.
3	Write in the third person.	"I believe..." in the passage becomes "The writer believes..." or simply "It is argued that...". Never carry "I", "we", "you" across.
4	Write in the past tense — for narrative and reported opinion.	Universal truths and permanent facts stay in the present ("Water boils at 100°C"; "The Himalayas lie to the north").
5	Use your own words.	Some technical terms must be reused (you cannot paraphrase "photosynthesis"). But do not lift whole sentences.
6	No new material.	No examples of your own, no opinions of your own, no facts from outside the passage.
7	No direct speech, no quotations, no dialogue.	Convert to indirect speech.
8	One continuous paragraph.	A précis is never bulleted and rarely more than one paragraph.
9	Keep the author's proportion and order.	If a third of the passage is about causes, roughly a third of your précis is about causes.
10	No rhetorical questions, no exclamations.	Convert "Is this acceptable?" to "The writer questions whether this is acceptable."
11	Cut all: examples, illustrations, repetitions, figures of speech, digressions, adjectives of emphasis.	This is where your compression comes from.
12	State the word count at the end in brackets.	e.g. (118 words). Shows you obeyed rule 1.

3.3 Step-by-step method (the 25-minute procedure)

Step 1 — Read once, fast (2 min). Do not write. Just get the gist. Ask: *what is this passage arguing?*

Step 2 — Read again with a pencil (4 min).

- Underline the **topic sentence** of each paragraph.
- **Strike through** every example, every "for instance", every list of illustrations, every quotation, every repetition.
- Circle **connective words** (however, therefore, moreover) — they carry the logical skeleton and you must preserve the logic, if not the words.

Step 3 — Count the original (1 min).

Count the words in one full line, multiply by the number of lines. Divide by three.

Write your target on the rough sheet: e.g. **Target: 120 words (±12)**.

Step 4 — Write the skeleton in the margin (3 min).

Reduce each paragraph to one clause. Four paragraphs → four clauses. That is your précis in miniature.

Step 5 — Draft in rough (6 min). Join the clauses into flowing prose. Third person, past tense. Do not look back at the passage while drafting — this is what forces "own words".

Step 6 — Count and trim (3 min). Over the target? Cut adjectives first, then merge sentences with participles ("Having considered X, the writer argues Y").

Step 7 — Write the fair copy + title (5 min). Title last, once you know what the passage is really about. Add **(N words)**.

Step 8 — Run the checklist (1 min). See §3.6.

3.4 Compression techniques (how you actually get to one-third)

Technique	Before	After
Phrase → word	"in spite of the fact that"	"although"
List → category	"rice, maize, millet and wheat"	"cereals"
Clause → participle	"He finished the report and then submitted it."	"Having finished the report, he submitted it."
Drop the illustration	"Pollution is severe — in Delhi, for example, ..."	"Pollution is severe."
Two sentences → one	"The scheme failed. It lacked funds."	"The scheme failed for want of funds."
Drop intensifiers	"extremely important and highly significant"	"important"
Nominalise	"when the government decided to intervene"	"on government intervention"

3.5 Worked example

The passage (352 words)

Every year, in the hill districts of the North-East, the rains arrive with a ferocity that outsiders find difficult to imagine. Roads that took three years to build vanish in a single night. Culverts collapse. Villages that were a two-hour drive from the district headquarters become, for weeks at a time, effectively islands. I have watched a schoolteacher in one such village walk eleven kilometres along a ridge because the road below had simply ceased to exist.

The usual response is to blame the contractor. Sometimes that blame is deserved. But the deeper problem is that we have been building the wrong kind of road in the wrong kind of place. Engineering practice in India was largely codified for the plains, where the chief enemies are traffic load and temperature. In young, folded, heavily-rained-upon mountains, the chief enemy is water — water moving through soil, water undercutting slopes, water finding every seam in the rock. A road that is merely strong is not enough. A road must also drain.

There is, moreover, a question of who builds. For decades, hill roads have been designed in distant offices by engineers who have never walked the alignment in August. Local knowledge — which slope slips, which spring runs all year, where the old footpath ran and why it ran there — is treated as folklore rather than as data.

Yet villagers can often predict a landslide site with an accuracy that would embarrass a soil report.

The remedy is not more money alone, though more money is needed. It is a change of method: design that begins with drainage rather than ending with it; specifications written for mountains rather than borrowed from the plains; and a genuine role for the village in choosing and maintaining its own alignment. Where communities have been given charge of small assets — as some states in the region have experimented with — maintenance has often improved dramatically, because the asset belongs to somebody. Until that shift happens, we will keep rebuilding the same road, and the rain will keep taking it away.

Model précis (117 words — one-third of 352 ≈ 117)

Title: Why Hill Roads Fail

The writer observed that heavy monsoon rain in North-Eastern hill districts repeatedly destroyed newly built roads and cut villages off for weeks. While contractors were often blamed, he argued that the deeper fault lay in method: Indian engineering practice was framed for the plains, where load and temperature matter most, whereas in young, high-rainfall mountains the decisive threat is water. A road therefore had to drain as well as bear weight. He further noted that alignments were designed remotely by engineers unfamiliar with the terrain, and that villagers' knowledge of unstable slopes was dismissed. The remedy he proposed was drainage-led design, mountain-specific specifications, and community responsibility for choosing and maintaining roads.

(117 words)

What that précis did, move by move

- **Title** given, short, captures the argument (not just the subject).
- "I have watched a schoolteacher..." — a pure illustration — **deleted entirely**.
- First person "I", "we" → **"the writer"**, third person throughout.
- Past tense for the writer's reported argument ("argued", "noted", "proposed"); but the general truth "the decisive threat is water" kept in the present. Both are correct.
- Order of the original (problem → misdiagnosis → real cause → who builds → remedy) **preserved exactly**.
- No new material: the "some states in the region" clause was generalised, not turned into a named example.
- Rhetorical richness ("the rain will keep taking it away") dropped — précis is plain.

3.6 The pre-submission checklist

Tick all nine before you move on:

- ☐ Title written and underlined
- ☐ Length within ±10% of one-third
- ☐ Word count stated in brackets
- ☐ Third person — no "I", "we", "you", "our"
- ☐ Past tense for reported argument; present only for permanent truths
- ☐ One continuous paragraph, no bullets, no headings
- ☐ No quotations, no direct speech, no rhetorical questions
- ☐ No examples, no illustrations, no repetition
- ☐ Nothing in it that was not in the passage

3.7 Practice plan

Find any 300–450 word passage — a newspaper editorial is ideal — and précis it in 25 minutes. **Do six of these across the 21 days.** Editorials from *The Hindu*, *Indian Express* or *Nagaland Post* work perfectly and double as current-affairs revision. Precision comes from repetition, not from reading about précis.

4. AMPLIFICATION — 10 marks

4.1 What amplification is (most candidates get this wrong)

You are given a **proverb, maxim, quotation or short saying** — for example "*A stitch in time saves nine*" or "*Knowledge is power*" — and asked to **amplify** it.

Amplification means: expand this compressed statement into a developed piece of prose that explains what it means, why it is true, and where it applies.

It is **not**:

- ✗ a mini-essay on a topic (you must stay tethered to the saying);
- ✗ a definition (too short);
- ✗ a story (an example is one *part* of it, not the whole);
- ✗ a précis (that is compression; this is expansion — the opposite).

Think of it as: *the saying is a seed; you grow it into a plant of five parts.*

4.2 Length and format

Words	200–300 (250 is the sweet spot)
Paragraphs	3–5 short paragraphs, or one well-signposted block
Time	20 minutes total
Opening	Quote the maxim itself in the first line, in quotation marks
Heading	Write the maxim as the heading, underlined

Do not write 600 words. Amplification is worth 10 marks — the same as grammar. Match your effort to the marks.

4.3 The five-move structure (memorise this)

Move	What you write	~Words
1. Literal meaning	What the words say on their face. Unpack the image or metaphor concretely.	40
2. Deeper / general meaning	The universal principle behind the image. This is the heart of the answer.	60
3. Illustration	One or two concrete examples — historical, everyday, scientific, or from public life. This is where most candidates are weakest and where marks are easiest.	70
4. Contemporary relevance	Why this matters now — link to modern life, technology, governance, society, India today.	50
5. Conclusion	Restate the maxim in your own words, elevated. One or two sentences.	30

The single most useful tip: move 3 (illustration) is what separates a 5/10 from an 8/10. Have a stock of examples ready that can be bent to almost any maxim — see §4.5.

4.4 Worked example 1 — "A stitch in time saves nine"

"A stitch in time saves nine"

Taken literally, the proverb describes a piece of cloth in which a single thread has given way. If the small tear is mended at once, one stitch is enough. If it is ignored, the tear widens with every use until nine stitches, or a new garment altogether, are required. The image is domestic, but the logic is universal.

The deeper meaning is that problems have a cost curve that rises with delay. Difficulty is rarely static; neglect compounds it. A small defect attended to early consumes a little effort at a moment when effort is cheap. The same defect left alone draws in other systems, hardens into habit, and eventually demands resources far beyond what timely action would have required. Promptness, the proverb suggests, is not merely a virtue of character but a form of economy.

Illustrations are everywhere. In medicine, a condition detected at screening is often curable, while the same condition detected late may be untreatable at any price — which is precisely the argument for preventive health programmes. In engineering, a hairline crack in a bridge or a small unpatched flaw in software costs little to repair and a great deal to ignore; most large system failures begin as a small fault that nobody thought worth attending to. In public administration, a minor local grievance addressed within a week may never be heard of again, while the same grievance ignored for a year can become a movement.

The saying has gained rather than lost force in our own time. Modern systems are tightly coupled, so a small failure propagates faster than it once did. Climate change is perhaps the largest example of a stitch not taken in time: emissions that could have been curbed cheaply decades ago now demand a far costlier global effort. The same holds in personal life, where a postponed examination syllabus or an unattended savings habit exacts its price later.

In short, the proverb counsels vigilance rather than mere haste. To act early is not to act anxiously; it is to recognise that time itself is an ingredient in the cost of every problem. The stitch taken today is the cheapest stitch we will ever take.

(≈340 words — trim the third paragraph to two examples for a 250-word version.)

4.5 Worked example 2 — "Knowledge is power"

"Knowledge is power"

The phrase, associated with Francis Bacon, sets two words side by side that at first seem unrelated. Literally it claims that knowing something — how a thing works, what the law says, where the money went — confers the capacity to act upon it. Power here is not force but the ability to bring about intended effects.

At a deeper level the maxim describes the relationship between information and control. The person who understands a system can predict it; the person who can predict it can shape it. This is why, historically, knowledge has been guarded: priesthoods restricted scripture, guilds restricted craft secrets, and colonial administrations restricted technical education. Conversely, every genuine expansion of freedom in history has been accompanied by an expansion of literacy. Ignorance is not a neutral absence; it is a condition in which one is acted upon rather than acting.

The illustrations are striking. The scientific revolution converted knowledge of

nature into power over nature — medicine, electricity, agriculture. In India, the Right to Information Act, 2005 rests on exactly this proposition: it gives the ordinary citizen access to official records, and in doing so shifts a measure of power from the office to the applicant. At an individual level, a farmer who knows the market price of his crop bargains differently from one who does not.

In the present century the maxim needs a qualification. We live amid an unprecedented abundance of information, yet abundance is not knowledge. Data must be verified, understood and applied before it becomes power; unverified information can instead be used to manipulate, as the spread of misinformation online demonstrates. Moreover, knowledge concentrated in a few hands — in large technology firms or in the state — produces power that is equally concentrated. The democratic task of our time is therefore not only to create knowledge but to distribute it.

Knowledge, then, is power only when it is true, understood, and shared. Held by the few it is dominance; held by the many it is freedom.

(≈330 words)

4.6 Worked example 3 — *"United we stand, divided we fall"*

"United we stand, divided we fall"

The literal picture is military. A body of people standing together presents an unbroken front and cannot easily be pushed over; the same people scattered are overcome one by one. The saying compresses a fact about physical formations into a rule about human affairs.

Its deeper claim is that collective strength is not simply the sum of individual strengths. Unity adds something — coordination, mutual support, shared purpose — that the same individuals acting separately do not possess. Division subtracts correspondingly, because a divided group spends against itself the energy it would otherwise spend on its common problem. This is why those who wish to weaken a community rarely attack it directly; they set its parts against one another.

History supplies both halves of the lesson. India's freedom struggle became irresistible only when it drew together regions, languages, faiths and classes into a single movement; the same subcontinent had earlier been subdued piecemeal because its kingdoms would not combine. The formation of cooperative societies, of trade unions, and of self-help groups follows the same principle at a smaller scale. In the North-East, the practical value of unity is visible in the village institutions that manage common forests, water sources and footpaths — assets no single household could maintain but which a community sustains without difficulty.

Today the maxim applies to problems that respect no boundary. Pandemics, climate change and cybercrime cannot be answered by any one state acting alone, which is why international cooperation, however imperfect, persists. Within a country, unity means something more demanding than uniformity: it means holding together across real differences of tribe, language and belief, and resisting the easy politics that profits from sharpening those differences.

Unity, properly understood, is therefore not the absence of difference but the presence of a shared purpose strong enough to carry difference. That is what allows a people to stand; its loss is what makes them fall.

(≈325 words)

4.7 A stock of illustrations you can bend to almost any maxim

Have these ready. In the hall you will be able to attach at least two to whatever maxim appears:

Domain	Reusable illustration
Science / medicine	Preventive screening vs late diagnosis; vaccination; the scientific method as patient accumulation
Technology	A small software bug that brings down a large system; the digital divide; misinformation online
Indian history	The freedom struggle as unity across differences; Gandhi's insistence on means as well as ends; social reform movements
Governance	The Right to Information Act, 2005; MGNREGA as a right rather than a favour; decentralisation through Panchayati Raj
Nagaland / North-East	Village councils and community management of common resources; the Communitisation Act, 2002; jhum cultivation and the cost of short-term choices; Hornbill Festival and cultural unity in diversity
Environment	Climate change as delayed action; deforestation; the tragedy of the commons
Everyday life	Studying steadily vs cramming; savings and compound interest; maintenance of a house or a vehicle

4.8 Fifteen maxims most likely to appear

Prepare a 3-line mental plan for each — literal, deeper, one example.

1. A stitch in time saves nine
2. Knowledge is power
3. United we stand, divided we fall
4. Honesty is the best policy
5. Where there is a will, there is a way
6. Necessity is the mother of invention
7. Health is wealth
8. Time and tide wait for no man
9. Look before you leap
10. Actions speak louder than words
11. A friend in need is a friend indeed
12. All that glitters is not gold
13. Failure is the pillar of success
14. Simple living, high thinking
15. Service to man is service to God

4.9 Amplification checklist

- [] Maxim written as the heading, underlined
- [] Maxim quoted in the first sentence
- [] Literal meaning stated before the deeper meaning
- [] At least **two** concrete illustrations
- [] Explicit contemporary/Indian relevance
- [] Conclusion restates the maxim in fresh words
- [] 200–300 words, not 600
- [] Did not drift into a general essay on the topic

5. CONSTRUCTION OF SENTENCE & GRAMMAR — 10 marks

Cheapest marks on the paper. Target **9/10**. This section is pure recall and pattern recognition, which is exactly what an engineer's brain is good at. Two hours of study here is worth more than ten hours on the essay.

5.1 Articles (a / an / the)

The rule in plain English: **a/an** = one of many, first mention, unspecified.

the = we both know which one. No article = general/uncountable/plural concepts.

Rule	Example
a/an before a vowel sound , not a vowel letter	an hour, an MP, an honest man but a university, a European, a one-rupee coin
the before superlatives, ordinals, unique things	the best, the first, the sun, the Constitution
the before rivers, seas, mountain ranges , groups of islands	the Ganga, the Himalayas, the Andamans but Mount Everest (single peak)
the before nationalities/adjectives used as nouns	the rich, the poor, the Naga people
No article before abstract nouns, meals, languages, most countries	Honesty pays. He had lunch. She speaks English. He went to India.
the for a class taken as a whole	The tiger is a wild animal.
the before names of newspapers, ships, holy books	the Hindu, the Titanic, the Bible
No article with "go to school/college/hospital/bed/prison" for their purpose	He went to school. but His father went to the school to meet the teacher.

5.2 Prepositions — the high-frequency traps

Correct	Not
married to	married with
discuss (no preposition)	discuss about
comprise / consist of	comprised of (<i>disputed</i> — <i>safest</i> : "comprises" or "consists of")
congratulate on	congratulate for
die of (a disease) / die from (injury) / die by (violence)	—
prefer X to Y	prefer X than Y
superior to / inferior to / senior to / junior to	superior than
different from	different than (<i>AmE</i>)
capable of , incapable of	capable to
insist on , depend on , rely on	insist to
suffer from	suffer with
familiar with (a thing) / familiar to (a person)	—

5.4 Tenses — the 12 forms and what each is for

	Simple	Continuous	Perfect	Perfect Continuous
Past	I wrote	I was writing	I had written	I had been writing
Present	I write	I am writing	I have written	I have been writing
Future	I shall/will write	I shall be writing	I shall have written	I shall have been writing

High-yield tense rules:

Rule	Example
Since / for + present perfect continuous for something still going on	He has been living here since 2019. (not "is living since")
Completed action with a past time marker → simple past	I saw him yesterday. (not "have seen him yesterday")
Two past actions → the earlier one takes past perfect	The train had left before I reached the station.
After " if " (2nd conditional) → past subjunctive + would	If I were you, I would accept.
Conditional type 1	If it rains , we shall stay in.
Conditional type 3	If he had studied , he would have passed .
No future tense after when/if/until/before/as soon as	I will call you when he comes . (not "will come")
Universal truths → present simple always	The teacher said that the earth moves round the sun.
" It is time / I wish " → past tense form	It is time we went home. I wish I knew the answer.
Verbs of state (know, believe, own, seem, contain, love) → not continuous	I know him. (not "I am knowing him")

5.5 Active and passive voice

Conversion rule: the **object** of the active sentence becomes the **subject** of the passive; the verb becomes **be + past participle**; the old subject goes into a **by** phrase (often dropped).

Tense	Active	Passive
Present simple	He writes a letter.	A letter is written by him.
Present continuous	He is writing a letter.	A letter is being written by him.
Present perfect	He has written a letter.	A letter has been written by him.
Past simple	He wrote a letter.	A letter was written by him.
Past continuous	He was writing a letter.	A letter was being written by him.
Past perfect	He had written a letter.	A letter had been written by him.
Future simple	He will write a letter.	A letter will be written by him.
Future perfect	He will have written a letter.	A letter will have been written by him.

Modal	He can write a letter.	A letter can be written by him.
Imperative (order)	Shut the door.	Let the door be shut .
Imperative (request)	Please help me.	You are requested to help me.
Question (do)	Did he write it?	Was it written by him?
Question (wh-)	Who wrote it?	By whom was it written?
Infinitive	I have a job to do.	I have a job to be done .

Note: perfect continuous and future continuous have no natural passive. If asked, the question is faulty — don't panic, rephrase.

5.6 Direct and indirect speech

Three things change: the **tense** (back-shift), the **pronouns**, and the **words of time and place**.

Reporting verb in the past → back-shift:

Direct	Indirect
present simple	past simple
present continuous	past continuous
present perfect	past perfect
past simple	past perfect
will	would
can	could
may	might
must	had to / must
shall	should / would

No back-shift if: the reporting verb is in the present/future, or the statement is a universal truth, or a historical fact.

Time & place words:

Direct	Indirect
now	then
today	that day
tomorrow	the next day / the following day
yesterday	the previous day / the day before
last night	the previous night
next week	the following week

here	there
this	that
these	those
ago	before
thus	so

Sentence-type conversions:

Type	Direct	Indirect
Statement	He said, "I am tired."	He said that he was tired.
Yes/no question	He said, "Are you ready?"	He asked if/whether I was ready.
Wh- question	He said, "Where do you live?"	He asked where I lived .
Command	He said, "Shut the door."	He ordered me to shut the door.
Request	He said, "Please help me."	He requested me to help him.
Advice	He said, "You should rest."	He advised me to rest .
Negative command	He said, "Don't go."	He forbade me to go / told me not to go .
Exclamation	He said, "What a fine day!"	He exclaimed that it was a very fine day.
Wish	He said, "May you succeed!"	He wished that I might succeed.

In questions, the reported clause takes **statement word order** — "he asked where I lived", never "he asked where did I live". This is the most common error.

5.7 Commonly confused words

Word	Meaning	Word	Meaning
accept	to receive	except	apart from
affect (v)	to influence	effect (n)	result
principal	chief / head of college	principle	a rule or belief
stationary	not moving	stationery	writing materials
complement	to complete	compliment	praise
council	an assembly	counsel	advice / a lawyer
advice (n)	guidance	advise (v)	to give guidance
practice (n)	—	practise (v)	—
its	belonging to it	it's	it is
your	belonging to you	you're	you are
lose	to misplace	loose	not tight

beside	next to	besides	in addition to
eminent	famous	imminent	about to happen
elicit	to draw out	illicit	illegal
later	afterwards	latter	the second of two
farther	more distance	further	additional
historic	important in history	historical	relating to history
respectable	worthy of respect	respectful	showing respect
credible	believable	creditable	praiseworthy
industrious	hardworking	industrial	of industry
economic	of the economy	economical	thrifty
continual	repeated	continuous	unbroken
momentary	brief	momentous	very important
canvas	cloth	canvass	to solicit votes
die	to cease living	dye	to colour
desert	arid land / abandon	dessert	sweet course
allusion	indirect reference	illusion	false impression
ingenious	clever	ingenuous	innocent, frank
luxurious	comfortable	luxuriant	growing abundantly
judicial	of the courts	judicious	showing good judgement
verbal	in words	verbose	wordy

5.8 Idioms and phrases (high-frequency)

Idiom	Meaning
a bolt from the blue	a sudden shock
at one's wits' end	completely perplexed
bury the hatchet	make peace
by leaps and bounds	very rapidly
call a spade a spade	speak plainly
carry the day	win
cut a sorry figure	make a poor impression
a fair-weather friend	one who deserts in trouble
get into hot water	get into trouble

Government by a few	oligarchy
Rule by the wealthy	plutocracy
A person who loves mankind	philanthropist
A person who hates mankind	misanthrope
One who eats human flesh	cannibal
One who lives on vegetables	vegetarian
A speech made without preparation	extempore
A life history written by oneself	autobiography
A life history written by another	biography
Something that cannot be corrected	incorrigible
Something that cannot be avoided	inevitable
Something that cannot be read	illegible
Something that cannot be heard	inaudible
Something that cannot be believed	incredible
Something that cannot be conquered	invincible
Words written on a tomb	epitaph
A place where money is coined	mint
A place for keeping birds	aviary
A place where bees are kept	apiary
A collection of maps	atlas
One who is present everywhere	omnipresent
One who knows everything	omniscient
One who is all-powerful	omnipotent
Fear of water	hydrophobia
Study of birds	ornithology
Study of insects	entomology
Study of earthquakes	seismology
Study of ancient things	archaeology
One who studies the stars scientifically	astronomer
One who claims to foretell by the stars	astrologer
A medicine that counteracts poison	antidote

A remedy for all diseases	panacea
One who deserts his religion	apostate
Occurring every year	annual
Occurring every two years	biennial
Occurring twice a year	biannual
Occurring every hundred years	centennial
One who talks too much	garrulous / loquacious
A person appointed to settle a dispute	arbitrator
A person who leaves his country to settle in another	emigrant (leaving) / immigrant (arriving)
One who is new to a trade	novice
The state of being unmarried	celibacy
One who does something for pleasure, not profession	amateur

5.10 Error spotting — the systematic method

Most error-spotting questions test one of only **eight** things. Check them in this order and you will find the error faster than by "reading for what sounds wrong":

1. **Subject–verb agreement** (find the true subject; ignore intervening phrases)
2. **Tense consistency** (any past marker? any two-past-actions structure?)
3. **Preposition** (is it on the trap list in §5.2?)
4. **Article** (missing **the** before a superlative? **a** before a vowel sound?)
5. **Pronoun** (does it agree in number? "Everyone...their" is loose; exams prefer "his")
6. **Parallelism** ("He likes reading, writing and **to swim**" → "and swimming")
7. **Redundancy** (return back, repeat again, more better, cope up with, discuss about, reason is because)
8. **Word choice / confusable** (§5.7)

5.11 Sentence transformation — the standard conversions

From	To	Example
Simple → Complex	insert a clause	<i>In spite of his poverty, he is happy.</i> → <i>Though he is poor, he is happy.</i>
Complex → Compound	use and/but/so	<i>Though he is poor, he is happy.</i> → <i>He is poor but he is happy.</i>
Affirmative → Negative	use "no...other / not...any"	<i>He is the best boy.</i> → <i>No other boy is as good as he.</i>
Assertive → Interrogative	rhetorical question	<i>Everybody wishes to be happy.</i> → <i>Who does not wish to be happy?</i>
Exclamatory → Assertive	—	<i>How beautiful the sky is!</i> → <i>The sky is very beautiful.</i>
Degrees: Positive → Comparative → Superlative	—	<i>No other city is as large as Mumbai.</i> → <i>Mumbai is larger than any other city.</i> → <i>Mumbai is the largest city.</i>

Too...to → so...that...not	—	<i>He is too weak to walk. → He is so weak that he cannot walk.</i>
Active → Passive	§5.5	—

5.12 Practice set (30 items) — answers at the end

A. Fill in the article (a / an / the / —)

1. He is ___ honest officer.
2. ___ Brahmaputra flows through Assam.
3. Kohima is ___ capital of Nagaland.
4. She is ___ university student.
5. ___ honesty is the best policy.

B. Fill in the preposition

6. He was congratulated ___ his success.
7. She is senior ___ me by two years.
8. They discussed ___ the matter for an hour. (*careful*)
9. He died ___ malaria.
10. I prefer tea ___ coffee.

C. Choose the correct verb

11. The Principal, along with the teachers, (was / were) present.
12. Neither the students nor the teacher (has / have) arrived.
13. The number of accidents (is / are) rising.
14. Physics (is / are) my favourite subject.
15. Ten kilometres (is / are) a long distance.

D. Correct the tense

16. He is living in Dimapur since 2018.
17. I have seen him yesterday.
18. The train had left before I have reached.
19. If I was you, I would resign.
20. I will call you when he will come.

E. Change the voice

21. The committee has approved the proposal.
22. Shut the window.
23. Who broke the glass?

F. Change to indirect speech

24. He said to me, "Where are you going?"
25. She said, "I will finish the work tomorrow."

G. Spot the error (the underlined-style part is in brackets)

26. (Each of the boys) (have) (submitted) (their assignment).
27. (He is one) (of those men) (who) (is never satisfied).
28. (The reason) (why he failed) (is because) (he did not study).

H. One-word substitution

29. A government run by officials.
30. One who can speak many languages.

ANSWER KEY — Section 5.12 (click to expand)

1. **an** (silent 'h' → vowel sound)
2. **The** (rivers take "the")
3. **the** (unique — "the capital of")
4. **a** ("university" begins with a /ju:/ consonant sound)
5. **—** (no article before an abstract noun used generally)

6. **on** — congratulate **on** 7. **to** — senior **to** (Latin comparatives take "to", never "than") 8. **—** (no preposition; "discuss the matter". "Discuss about" is wrong.) 9. **of** — die **of** a disease 10. **to** — prefer X **to** Y 11. **was** — "along with" does not make the subject plural 12. **has** — with neither...nor, the verb agrees with the nearer subject ("teacher") 13. **is** — "the number of" is singular 14. **is** — Physics is singular despite the -s 15. **is** — a single quantity of distance 16. He **has been living** in Dimapur since 2018. 17. I **saw** him yesterday. 18. The train had left before I **reached**. 19. If I **were** you, I would resign. 20. I will call you when he **comes**. 21. The proposal **has been approved** by the committee. 22. **Let the window be shut.** 23. **By whom was the glass broken?** 24. He **asked me where I was going.** 25. She said **that she would finish the work the next day.** 26. **have → has** ("Each of the boys **has**"). Also "their" is loose; "his" is preferred. 27. **is → are** ("one of those men **who are** never satisfied" — "who" refers to "men") 28. **is because → is that** ("The reason ... is that he did not study." "Reason ... because" is redundant.) 29. **bureaucracy** 30. **polyglot**

5.13 How to study this section in 21 days

- **Days 1–4:** read §5.1–5.7 once each day. Do not memorise — just re-read. Familiarity is enough for 80% of these.
- **Days 5–14:** 20 minutes a day on §5.8–5.9 (idioms and one-word substitutions). These are pure vocabulary and need spaced repetition. Make 5 flashcards a day.
- **Days 15–20:** do the practice set again from cold, plus any grammar questions from the PYQ folder. Target: under 15 minutes, 27+/30.
- **Day 21:** skim the tables only. No new material.

6. GENERAL KNOWLEDGE — 50 marks

Half the paper. This is where 21 days of study actually move the needle.

6.1 The seven areas and how to prioritise them

The official wording lists: **current events, general science, mental ability, Indian history, Indian polity, Indian economy, Geography**. Assume roughly equal weight (~7 marks each) unless the PYQs in your **PYQ/** folder say otherwise — check them.

Priority for **a technical candidate with 21 days**:

Rank	Area	Why this rank	Days to give it
1	Mental Ability	You are an engineer. This is arithmetic and logic. You can go from 0 to near-full marks in ~2 days because the question types are finite and mechanical. Highest marks per hour on the entire paper.	2
2	Indian Polity	Small, closed, highly predictable syllabus. Constitution facts repeat year after year. And Article 371A is near-certain in a Nagaland state exam .	3
3	Nagaland / North-East GK	Not named separately in the syllabus, but a state PSC <i>always</i> embeds state questions inside history/geography/polity/current events. Almost nobody prepares this systematically. Highest-yield, least-competed.	3
4	General Science	You already know most of the physics and chemistry. Only biology needs real work (human body, diseases, vitamins, nutrition).	2

5	Geography	Learnable in tables — rivers, ranges, soils, crops, minerals. Plus Nagaland geography, which overlaps with #3.	2
6	Indian Economy	Concept-heavy but the exam-level questions are shallow: inflation, GDP, RBI, budget, schemes.	2
7	Indian History	The largest syllabus with the lowest return per hour. Restrict yourself to modern India (1857–1947) plus a thin skim of ancient/medieval. Do not attempt to master it.	3
8	Current Affairs	Cannot be crammed, but a targeted sweep of the last 12 months pays. Do it in short daily doses, not one block.	30 min daily

6.2 The 21-day GK schedule (fits 3–4 hours/day alongside English)

Days	GK focus (≈2 hrs/day)	English focus (≈1.5 hrs/day)
1–2	Mental Ability — all 9 question types + practice	Read essay playbook + 4 model essays
3–5	Indian Polity incl. Article 371A	Grammar §5.1–5.7; 1 timed essay
6–8	Nagaland & North-East GK (whole capsule section)	1 précis + 1 amplification
9–10	General Science (biology emphasis)	Grammar §5.8–5.9 idioms/OWS; 1 timed essay
11–12	Geography (India + Nagaland)	1 précis + 1 amplification
13–14	Indian Economy	1 timed essay (Nagaland topic)
15–17	Indian History (modern-heavy)	1 précis + 1 amplification; grammar practice set
18–19	Current affairs sweep + revise weakest two areas	1 timed essay; quotation bank
20	Full 3-hour mock of the whole paper	(part of the mock)
21	Light revision only — tables, capsule, quotations. Sleep early.	Skim only

Every day, regardless: **30 minutes of current affairs** (see [03_GK_CAPSULE.md](#) §8).

6.3 How to answer "short descriptive" GK questions

This is the format most technical candidates handle badly. A short-descriptive GK question is typically 3 or 5 marks and asks something like:

"Write a short note on Article 371A of the Indian Constitution." (5)
"What is inflation? Briefly explain its causes." (3)
"Mention any four features of the Preamble." (4)

The universal template:

[Definition / what it is — 1 sentence]
 [Key facts, as 3–5 short numbered or bulleted points]

[Significance / why it matters – 1 sentence]

Rules of thumb:

Marks	Words	Structure
2	30–40	One definition sentence + two facts
3	50–70	Definition + 3 bullets
5	90–130	One-line definition + heading-led bullets + a closing significance line
8–10	200–250	Short intro ¶ + 3 sub-headings + conclusion

Do:

- **Use the exact keyword the examiner is scanning for.** For Art. 371A, the words "religious or social practices of the Nagas", "customary law and procedure", "administration of civil and criminal justice", "ownership and transfer of land and its resources", "resolution of the Legislative Assembly" are the marks. Get them in.
- **Use bullets and sub-headings** for anything 4 marks and above. Examiners marking 200 scripts find bullets faster than prose, and speed of finding = marks.
- **Underline the key term once** in your answer.
- **Answer the exact number asked.** "Any four features" → write exactly four, clearly numbered. Writing seven wastes time and can invite a penalty if some are wrong.

Don't:

- Don't write an essay. A 3-mark question answered in 300 words costs you a 5-mark question elsewhere.
- Don't waffle before the definition. Start with the answer in the first line.
- Don't leave anything blank. A partially remembered fact scores more than a blank.

Worked example — a 5-mark answer:

Q. Write a short note on Article 371A. (5)

Ans. Article 371A of the Constitution contains special provisions for the State of Nagaland. It was inserted by the **Constitution (Thirteenth Amendment) Act, 1962**, giving effect to the 16-Point Agreement between the Government of India and the Naga People's Convention, and came into force with Nagaland's statehood on 1 December 1963.

Its main features are:

1. **No Act of Parliament** shall apply to Nagaland in respect of —
(a) religious or social practices of the Nagas, (b) Naga customary law and procedure, (c) administration of civil and criminal justice involving decisions according to Naga customary law, and (d) ownership and transfer of land and its resources — **unless the Legislative Assembly of Nagaland so decides by a resolution.**
2. The **Governor of Nagaland** has special responsibility for law and order in the State so long as internal disturbances continue, exercising his individual judgement after consulting the Council of Ministers.
3. It provides for a **Regional Council for the Tuensang district** and special transitional arrangements for that area.

Significance: Article 371A constitutionally protects Naga customary law, land rights and social practices, and is regarded as the cornerstone of Nagaland's

distinct political and legal identity within the Indian Union.

Notice: definition first, the amendment number and year, the four protected subjects verbatim, the Governor's role, and a one-line significance. That is a full-marks answer and it takes four minutes to write.

6.4 Objective GK — tactics

- **No stated negative marking on this paper.** (Negative marking is specified only for the Technical MCQ paper.) So **attempt every single question.**
- Eliminate first. Two options are usually obviously wrong; a 50/50 guess is worth taking.
- Beware "All of the above" — it is correct more often than chance, but only when you can verify at least two options.
- Extremes ("always", "never", "only") in an option usually signal a wrong option.
- Numbers: if three options cluster and one is an outlier, the answer is usually in the cluster.
- **Do not go back and change an answer** unless you have a positive reason. First instinct on factual recall is right more often than second-guessing.

7. The final 72 hours

Day 19 (24 Sept): Full 3-hour mock. Same start time as the exam (9:00 am) if you can. Write it by hand, in one sitting, with a real clock. This is the single most valuable study session in the entire 21 days — it calibrates your speed, which is the only thing you cannot learn from a book.

Day 20 (25 Sept):

- Morning: review the mock. Note exactly where you overran time.
- Afternoon: revise the GK capsule tables only — Polity, Nagaland, Geography.
- Evening: re-read the essay skeletons, the précis rules, the amplification 5-move structure. Read the quotation bank once.
- Pack: admit card, photo ID, pens (3 of the same colour), spare refills, watch, transparent water bottle. Check the venue location and travel time.
- **Sleep by 22:00.** No new material after 20:00.

Day 21 (26 Sept — exam day):

- Light breakfast. Nothing heavy or unfamiliar.
- Arrive 45 minutes early.
- First 60 seconds in the hall: **read the essay topics and pick one.** Then start with grammar.
- Write your three clock checkpoints (§1.4) on the rough sheet before you write anything else.
- If you blank on a GK question, move on immediately. There are fifty marks of GK and none of them are worth two minutes of staring.

8. One-page cheat sheet (copy this out by hand on Day 20)

ORDER: Grammar(15) → GK-obj(25) → GK-desc(35) → Amplification(20)
→ Précis(25) → Essay(50) → Buffer(10)

CHECKPOINTS: 10:00 grammar done | 11:15 GK done | 11:40 essay starts

ESSAY 800-1000 words, 6-9 ¶, 8 min plan first.
Intro = hook / context / thesis / roadmap
Body = PESTLE dimensions + past-present-future
ALWAYS add one Nagaland / North-East paragraph
Concl = synthesise / restate thesis / conditional forward look
Never "Since time immemorial..."

PRÉCIS 1/3 length | TITLE | 3rd person | past tense | own words
no examples | no quotes | one paragraph | state word count

AMPLIF. 250 words | 5 moves:
literal → deeper → 2 illustrations → today → restate
Quote the maxim in line 1. Heading = the maxim.

GRAMMAR check order: S-V agreement → tense → preposition → article
→ pronoun → parallelism → redundancy → confusables

GK Answer format: definition line → 3-5 bullets → significance line
3 mk = 60 words | 5 mk = 120 words
Attempt EVERYTHING — no negative marking on this paper
Art. 371A is near-certain. Know it cold.

PRESENTATION: margins, para breaks, underline the title only,
one ink colour, single-line strikeouts, number every answer.

9. Where the rest of the material lives

File	Contains
02_ESSAY_BANK.md	30 likely topics grouped by theme, 4 full model essays with annotations, 20 quotations, opening/closing sentence patterns
03_GK_CAPSULE.md	Polity (incl. Art. 371A in depth), History, Nagaland & North-East GK , Economy, Geography, General Science, Mental Ability with worked shortcuts, Current-affairs revision checklist
_Shared/ OFFICIAL_SYLLABUS.md	The official pattern — ground truth
PYQ/	Past papers — check these for the actual GK weighting . Owned by another process; read only.

Accuracy note: facts in this playbook that you intend to quote in the exam — particularly amendment numbers, dates and article numbers — should be cross-checked once against a standard reference (Laxmikanth for Polity, or the Constitution text itself for Article 371A). Anything marked *verify* in these files has not been confirmed against a live source.

● Core — number systems

Source: `_Shared/CORE_01_DigitalLogic.md`

A1. Positional number systems

Concept — plain English first

Every number system you will ever meet in this exam works the same way, and if you internalise *one* idea you never have to memorise a conversion rule again.

A number is written as a string of digits. **Each digit's position gives it a weight, and the weights are powers of the base.** In decimal (base 10) the weights going left from the decimal point are 1, 10, 100, 1000 — that is 10^0 , 10^1 , 10^2 , 10^3 . Going right they are 10^{-1} , 10^{-2} , 10^{-3} . That's all "positional notation" means. Change the base to 2 and the weights become 1, 2, 4, 8, 16...; change it to 16 and they become 1, 16, 256, 4096. So the *value* of any number is just "each digit multiplied by its weight, all added up":

$$N = d_{(n-1)} \cdot b^{(n-1)} + \dots + d_1 \cdot b^1 + d_0 \cdot b^0 + d_{(-1)} \cdot b^{(-1)} + \dots$$

Once you see it this way, **converting anything to decimal is one formula**, and the only thing you have to learn separately is how to go the *other* way (decimal $\rightarrow$ base b), which is repeated division for the integer part and repeated multiplication for the fraction.

Formal definition

A **positional number system of radix (base) b** represents a number as an ordered string of digits from the digit set $\{0, 1, \dots, b-1\}$, where the digit in position i (counting from 0 at the radix point, increasing leftward) carries weight b^i .

Key points

Base	Name	Digits	Digits needed per 1 hex digit	Common use
2	Binary	0,1	—	Machine level, logic circuits
8	Octal	0–7	1 octal = 3 binary bits	UNIX file permissions
10	Decimal	0–9	—	Human
16	Hexadecimal	0–9, A–F	1 hex = 4 binary bits	Memory addresses, MAC, colour, forensic hex dumps

- Hex letters: **A=10, B=11, C=12, D=13, E=14, F=15**.
- With n digits in base b you can represent **b^n** distinct values, range **0 to b^n-1** .
- 4 bits = 1 **nibble**; 8 bits = 1 **byte**; a **word** is machine dependent (16/32/64 bits).
- Largest n -bit unsigned value = $2^n - 1$. (8-bit $\rightarrow$ 255, 16-bit $\rightarrow$ 65535, 32-bit $\rightarrow$ 4294967295.)
- To store a decimal number of D digits you need approximately **$D \times 3.32$** bits (since $\log_2 10 \approx 3.3219$).

Powers of 2 — memorise this table cold

2^0	2^1	2^2	2^3	2^4	2^5	2^6	2^7	2^8	2^9	2^{10}
-------	-------	-------	-------	-------	-------	-------	-------	-------	-------	----------

Technique 3 — Binary ↔ Octal / Hex: grouping (no arithmetic needed)

Because $8 = 2^3$ and $16 = 2^4$, each octal digit is exactly 3 bits and each hex digit exactly 4 bits. **Group from the radix point outwards**, padding the outer ends with zeros.

Worked example 6. $(1101011.1011)_2 \rightarrow$ octal and hex

```
OCTAL — groups of 3 from the point:
001 101 011 . 101 100      (padded: left "00", right "0")
 1   5   3 .   5   4      → (153.54)8

HEX — groups of 4 from the point:
0110 1011 . 1011
 6    B .    B      → (6B.B)16
```

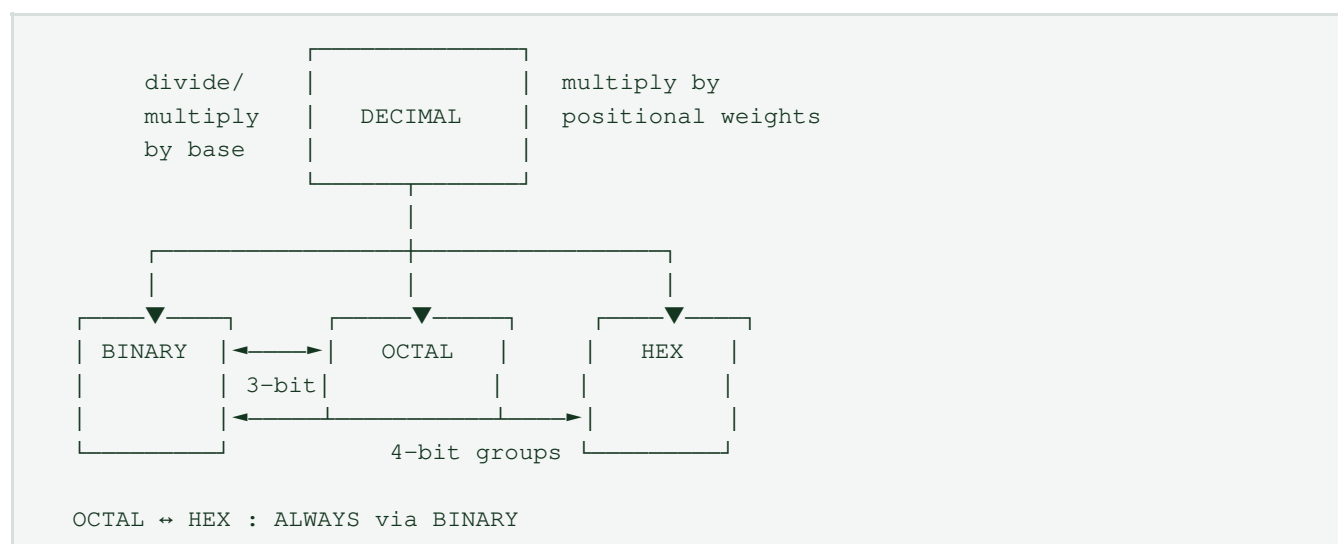
Worked example 7. $(3A7)_{16} \rightarrow$ binary $\rightarrow$ octal

```
3 = 0011, A = 1010, 7 = 0111 → 0011 1010 0111
Regroup in 3s from the right: 001 110 100 111 → (1647)8
```

Technique 4 — Octal ↔ Hex: always route through binary

Never try to convert octal to hex directly. 8 and 16 are not powers of each other in a convenient way. Go octal $\rightarrow$ binary $\rightarrow$ regroup $\rightarrow$ hex. (Worked example 7 shows the pattern in reverse.)

The full conversion map



Likely exam questions — Number systems

Q	Marks
Convert $(2AF.C)_{16}$ to binary, octal and decimal.	5
Convert $(0.6875)_{10}$ to binary and $(357.24)_8$ to hexadecimal, showing all steps.	5
Explain positional number systems. Convert $(1101.011)_2$ to decimal, $(89)_{10}$ to hex, and $(5C)_{16}$ to octal.	10

Discuss the four number systems used in digital computers, with conversion procedures between each pair, illustrating with examples. Why is hexadecimal preferred for representing memory addresses?

15

MCQ traps — Number systems

Trap	The right answer
"Read the remainders top-to-bottom"	Bottom-to-top for integer division; top-to-bottom for fraction multiplication. Examiners set the reversed string as a distractor.
Grouping binary from the <i>left</i>	Group from the radix point outwards . For integers that means from the right . $(1011)_2$ in octal is 001
Largest 8-bit number	Unsigned 255 (2^8-1), not 256. 256 is the <i>count</i> of values.
Number of distinct values in n bits	2^n values, range $0 \dots 2^n-1$.
$(11)_2$ vs $(11)_8$ vs $(11)_{16}$	3, 9, 17 respectively. Read the subscript.
"Octal to hex directly by grouping"	False. Must go via binary.
Is 1010 a valid BCD digit?	No — see A3.

A3. BCD (Binary Coded Decimal)

Concept

Pure binary is great for machines and horrible for humans reading a 7-segment display or a cash register. **BCD is a compromise: encode each decimal digit separately in 4 bits.** So 47 is *not* 101111 (pure binary); it is **0100 0111** — a "4" nibble followed by a "7" nibble. The price you pay is wasted space (6 of the 16 nibble patterns are illegal) and awkward arithmetic. The benefit is trivially easy conversion to and from a decimal display.

Key points

- The standard BCD code is **8421 BCD** (the weights of the four bits).
- Codes **1010, 1011, 1100, 1101, 1110, 1111 (10–15) are INVALID / forbidden** in BCD. Only 0000–1001 are legal. **Six unused codes** — very common one-mark MCQ.
- Unpacked BCD**: one decimal digit per byte (upper nibble is 0 or a zone). **Packed BCD**: two decimal digits per byte. Packed is twice as space-efficient.
- BCD is **less efficient** than pure binary: 4 bits carry only 10 of a possible 16 values, so BCD storage is roughly **20% larger** than binary for the same range.
- BCD is **not** the same as binary. $(25)_{10} = (11001)_2$ but $= (0010\ 0101)_{\text{BCD}}$.

Worked example 8 — BCD addition with the +6 correction

Rule: add the BCD digits as ordinary 4-bit binary. **If a nibble result exceeds 1001 (9), or if a carry out of that nibble occurred, add 0110 (6) to that nibble** and propagate the carry to the next digit.

Why +6? Because the nibble is naturally a mod-16 counter but BCD needs mod-10. You have to skip over the six illegal codes, so you push the value forward by 6.

Add 47 + 38 in BCD:

```
  0100 0111    (47)
+ 0011 1000    (38)
-----
  0111 1111    → low nibble 1111 = 15 > 9, INVALID → add 0110
      1111
    + 0110
    -----
    1 0101    → low digit = 0101 (5), carry 1 into high nibble
  high: 0111 + 1 = 1000 (8) → valid, no correction needed
-----
 1000 0101    = 85 ✓
```

Add 9 + 8 in BCD (single digit, generates a carry):

```
1001 + 1000 = 1 0001 → carry out of nibble occurred → add 0110 to the sum nibble
0001 + 0110 = 0111
Result: 0001 0111 = 17 ✓
```

MCQ traps — BCD

- How many 4-bit codes are unused in BCD? → 6.
- What is added for BCD correction? → 6 (0110), not 10.
- $(128)_{10}$ in BCD requires **12 bits** (3 digits $\times$ 4), in pure binary only 8.
- BCD 0001 0000 = **10**, not 16.

A4. Excess-3, Gray code, and other codes

Excess-3

Concept: Take the BCD code and add 3 to every digit. Sounds pointless — but it makes the code **self-complementing**: the 9's complement of a decimal digit is obtained by simply inverting all four bits (1's complement). That property made subtraction trivial in early decimal computers.

Decimal	BCD (8421)	Excess-3	Gray
0	0000	0011	0000
1	0001	0100	0001
2	0010	0101	0011
3	0011	0110	0010
4	0100	0111	0110
5	0101	1000	0111

6	0110	1001	0101
7	0111	1010	0100
8	1000	1011	1100
9	1001	1100	1101

- Excess-3 is **unweighted** and **self-complementing**. So are 2421 and 84-2-1.
- Verify self-complementing: 4 → 0111; invert → 1000 = Excess-3 for 5; and 9-4 = 5. ✓

Gray code (reflected binary code)

Concept: In Gray code, **consecutive values differ in exactly one bit**. Why care? Think of a rotary shaft encoder. If the position goes from 0111 to 1000 in pure binary, all four bits change at once, and if the sensors are a hair out of alignment you can momentarily read a wildly wrong value like 1111. With Gray code only one bit ever changes, so the worst error is off-by-one. Gray code is also the ordering used along the axes of a **Karnaugh map** — that is precisely why adjacent K-map cells differ in one variable.

Binary → Gray: keep the MSB; each subsequent Gray bit = XOR of adjacent binary bits.

$$g_{n-1} = b_{n-1} , \quad g_i = b_{i+1} \oplus b_i$$

Gray → Binary: keep the MSB; each subsequent binary bit = previous *binary* bit XOR current Gray bit.

$$b_{n-1} = g_{n-1} , \quad b_i = b_{i+1} \oplus g_i$$

Worked example 9. Binary 1011 → Gray

```

b3 b2 b1 b0 = 1 0 1 1
g3 = b3           = 1
g2 = b3 ⊕ b2 = 1⊕0 = 1
g1 = b2 ⊕ b1 = 0⊕1 = 1
g0 = b1 ⊕ b0 = 1⊕1 = 0
Gray = 1110

```

Back: Gray 1110 → binary

```

b3 = g3           = 1
b2 = b3 ⊕ g2 = 1⊕1 = 0
b1 = b2 ⊕ g1 = 0⊕1 = 1
b0 = b1 ⊕ g0 = 1⊕0 = 1
Binary = 1011 ✓

```

Other codes worth one line each

Code	Property
2421 (Aiken)	Weighted, self-complementing
84-2-1	Weighted (with negative weights), self-complementing

Biquinary	7 bits, 2 ones per code, error detecting
Parity (even/odd)	Adds 1 bit; detects any odd number of bit errors; cannot correct
Hamming code	Adds k parity bits where $2^k \geq m+k+1$; detects 2, corrects 1 bit error
Alphanumeric	ASCII, EBCDIC, Unicode

A5. ASCII, EBCDIC, Unicode

Key points

Code	Bits	Characters	Notes
ASCII	7 (often stored in 8 with parity/extension)	128	American Standard Code for Information Interchange
Extended ASCII	8	256	Adds box-drawing, accented characters
EBCDIC	8	256	IBM mainframes
Unicode	16-bit (UCS-2) / variable (UTF-8: 1–4 bytes)	65,536+ / >1,000,000 code points	UTF-8 is backward compatible with ASCII

Code points to memorise (they get asked directly):

Character	Decimal	Hex	Binary
NUL	0	00	0000000
Space	32	20	0100000
0	48	30	0110000
9	57	39	0111001
A	65	41	1000001
Z	90	5A	1011010
a	97	61	1100001
z	122	7A	1111010
DEL	127	7F	1111111

- **Difference between an uppercase and its lowercase letter is exactly 32 (0x20)** — i.e. only bit 5 differs. That's how a case-toggle is a single OR/AND with 0x20.
- Digits: ASCII value = digit + 48. So converting a character digit to its numeric value is `c - '0'`.
- ASCII codes 0–31 and 127 are **control characters** (non-printing). 32–126 are printable.

Forensic relevance (for the CF paper): hex dumps in forensic tools display file contents

as hex bytes plus an ASCII column. File signatures / magic numbers are read this way —
e.g. `FF D8 FF` = JPEG, `25 50 44 46` = %PDF, `50 4B 03 04` = ZIP/DOCX/XLSX,
`89 50 4E 47` = PNG. Being fluent in hex↔ASCII is a directly examinable forensic skill.

PART B — BINARY ARITHMETIC AND SIGNED NUMBERS

B1. Unsigned binary arithmetic

Addition and subtraction rules

ADDITION	SUBTRACTION
$0 + 0 = 0$	$0 - 0 = 0$
$0 + 1 = 1$	$1 - 0 = 1$
$1 + 0 = 1$	$1 - 1 = 0$
$1 + 1 = 0$ carry 1	$0 - 1 = 1$ borrow 1
$1 + 1 + 1 = 1$ carry 1	

Worked example 10. $1011 + 1101$

```
  1111    ← carries
  1011    (11)
+ 1101    (13)
-----
 11000    (24) ✓
```

Worked example 11. $1101 - 0111$ by direct borrow

```
  1101    (13)
- 0111    (7)
-----
  0110    (6) ✓
```

Multiplication and division

Worked example 12. 1101×101 ($13 \times 5 = 65$)

```
  1101
×   101
-----
  1101    ← ×1
  0000    ← ×0, shifted
 1101     ← ×1, shifted twice
-----
1000001    = 64 + 1 = 65 ✓
```

Binary multiplication is just **shift and add**: for each 1 bit in the multiplier, add a shifted copy of the multiplicand. This is exactly what a hardware multiplier does.

Worked example 13. $1011 \div 10$ ($11 \div 2 = 5$ remainder 1)

```
      101      ← quotient
10 ) 1011
    10
    --
     01
     00
     --
      11
      10
      --
       1      ← remainder
Quotient 101 = 5, remainder 1 ✓
```

D3. Decoders

Concept

A decoder turns a **coded input into a "one-hot" output**: exactly one of its output lines goes active for each input combination. Think of it as "which one of these 8 things did you name?" — this is precisely how a memory chip selects one word out of many from an address, and how an instruction opcode selects one control line in a CPU.

An **n-to-2ⁿ decoder** has n inputs and 2ⁿ outputs.

2-to-4 decoder with Enable

E	A1	A0	Y3	Y2	Y1	Y0
0	X	X	0	0	0	0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

$Y0 = E \cdot A1' \cdot A0'$ $Y1 = E \cdot A1' \cdot A0$
 $Y2 = E \cdot A1 \cdot A0'$ $Y3 = E \cdot A1 \cdot A0$

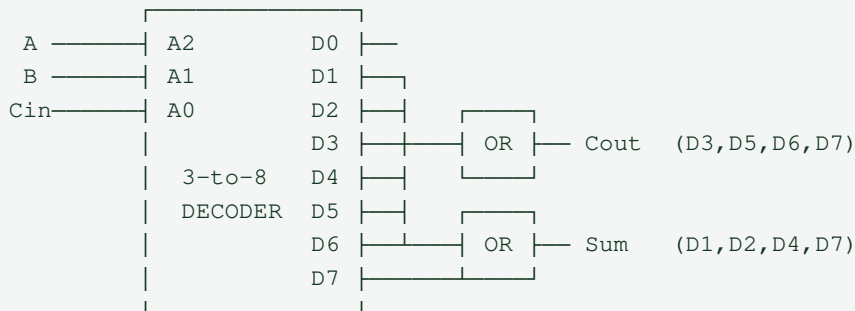
Notice: each output IS a minterm. That is the key to the next idea.

Implementing functions with a decoder

Because an n-to-2ⁿ decoder generates all 2ⁿ minterms, any function of n variables can be built as an OR of the appropriate decoder outputs.

Worked example 26. Implement a full adder using a 3-to-8 decoder.

Sum = $\Sigma m(1, 2, 4, 7)$ → OR together decoder outputs D1, D2, D4, D7
Cout = $\Sigma m(3, 5, 6, 7)$ → OR together decoder outputs D3, D5, D6, D7



If the decoder has **active-LOW** outputs (as real 74138 chips do), replace the OR gates with **NAND** gates. Very common exam refinement.

Decoder expansion

Two 3-to-8 decoders + one inverter make a **4-to-16 decoder**: use the MSB as the enable — directly to one chip, inverted to the other.

Encoders — the inverse

An encoder has **2ⁿ inputs and n outputs**; it outputs the binary code of whichever input line is active.

8-to-3 encoder (octal-to-binary):

$$\begin{aligned} Y2 &= D4 + D5 + D6 + D7 \\ Y1 &= D2 + D3 + D6 + D7 \\ Y0 &= D1 + D3 + D5 + D7 \end{aligned}$$

(Just OR together all input lines whose index has a 1 in that bit position.)

Two problems with a plain encoder — this is what gets examined:

- Ambiguity if two inputs are active at once** (D3 and D6 together would produce 111 = 7, which is wrong for both).
- Output 000 is ambiguous**: it means either "D0 is active" or "no input is active."

Fixed with an extra **valid/V** output.

Priority encoder solves problem 1: if several inputs are active it encodes the **highest-priority (usually highest-numbered)** one and ignores the rest.

4-to-2 priority encoder truth table (D3 highest priority):

D3	D2	D1	D0	Y1	Y0	V
0	0	0	0	X	X	0
0	0	0	1	0	0	1
0	0	1	X	0	1	1
0	1	X	X	1	0	1
1	X	X	X	1	1	1

$$Y1 = D3 + D2$$

$$Y0 = D3 + D2'D1$$

$$V = D3 + D2 + D1 + D0$$

Real-world use: interrupt priority resolution in a CPU (IC 74147 is a 10-to-4 priority encoder).

	Decoder	Encoder
Inputs → Outputs	$n \rightarrow 2^n$	$2^n \rightarrow n$
Direction	Code → one-hot	One-hot → code
Active outputs	Exactly one	The binary code
Typical use	Memory address selection, instruction decode, display drive	Keyboard scanning, interrupt priority

D5. Comparators, parity, code converters

Magnitude comparator

1-bit comparator:

$$A > B : A \cdot B'$$

$$A < B : A' \cdot B$$

$$A = B : A \odot B = A'B' + AB \quad (\text{XNOR} - \text{equality is XNOR})$$

n-bit comparison works from the MSB down: the numbers are equal only if **all** bit-pairs are equal (AND of all XNORs); $A > B$ if the highest-order differing bit has $A=1$.

For 4-bit:

$$(A = B) = x_3 \cdot x_2 \cdot x_1 \cdot x_0 \quad \text{where } x_i = A_i \odot B_i$$

$$(A > B) = A_3B_3' + x_3 \cdot A_2B_2' + x_3x_2 \cdot A_1B_1' + x_3x_2x_1 \cdot A_0B_0'$$

$$(A < B) = A_3'B_3 + x_3 \cdot A_2'B_2 + x_3x_2 \cdot A_1'B_1 + x_3x_2x_1 \cdot A_0'B_0$$

IC **7485** is a 4-bit magnitude comparator (cascadable).

Parity generator / checker

- **Even parity generator** for 3 data bits: $P = A \oplus B \oplus C$ (makes total number of 1s even).
- **Odd parity generator:** $P = (A \oplus B \oplus C)'$.
- **Checker:** XOR all bits including parity. Result 1 $\Rightarrow$ error (for even parity).
- Detects **any odd number** of bit errors; **cannot detect an even number** and **cannot correct** anything. IC 74180 is a 9-bit parity generator/checker.

Code converters

The general procedure: write a truth table with the **source code as inputs** and the **target code as outputs**, then K-map each output bit.

Binary → Gray (4-bit):

```

G3 = B3
G2 = B3 ⊕ B2
G1 = B2 ⊕ B1
G0 = B1 ⊕ B0

```

Three XOR gates. Very cheap.

Gray → Binary (4-bit):

```

B3 = G3
B2 = B3 ⊕ G2 = G3 ⊕ G2
B1 = B2 ⊕ G1 = G3 ⊕ G2 ⊕ G1
B0 = B1 ⊕ G0 = G3 ⊕ G2 ⊕ G1 ⊕ G0

```

Note this is a **cascade** (each output feeds the next) so it is slower.

BCD → Excess-3 (worked truth table, with 1010–1111 as don't-cares):

Decimal	B3 B2 B1 B0	E3 E2 E1 E0
0	0000	0011
1	0001	0100
2	0010	0101
3	0011	0110
4	0100	0111
5	0101	1000
6	0110	1001
7	0111	1010
8	1000	1011
9	1001	1100
10–15	—	X (don't care)

Minimising each output with a K-map (using the six don't-cares) gives the standard result:

```

E3 = B3 + B2B1 + B2B0          = B3 + B2 (B1 + B0)
E2 = B2'B1 + B2'B0 + B2B1'B0' = B2' (B1 + B0) + B2B1'B0' = B2 ⊕ (B1 + B0)
E1 = B1'B0' + B1B0             = B1 ⊙ B0      (XNOR)
E0 = B0'

```

Spot check, input 5 = 0101: E3 = 0 + 1·(0+1) = 1; E2 = 0 ⊕ (0+1) = ... **careful:** B2=1 so E2 = B2 ⊕ (B1+B0) = 1 ⊕ 1 = 0; E1 = B1 ⊙ B0 = 0 ⊙ 1 = 0; E0 = B0' = 0 → **1000** ✓ (Excess-3 for 5 is 1000).

BCD → 7-segment decoder: 4 inputs (BCD), 7 outputs (a–g). Segments are lit to form the digit. IC **7447** (common-anode, active-low outputs) / **7448** (common-cathode). Worth knowing the segment layout:



Digit 0 lights a,b,c,d,e,f (not g). Digit 1 lights only b,c. Digit 8 lights all seven.

Likely exam questions — Combinational circuits

Q	Marks
Distinguish between combinational and sequential circuits with examples.	5
Design a half adder and a half subtractor; give truth tables, expressions and logic diagrams.	5
Explain the difference between a decoder and a demultiplexer.	5
Design a full adder. Show how it can be built from two half adders and one OR gate.	10
Implement $F(A,B,C) = \Sigma m(1,2,6,7)$ using (i) a 3-to-8 decoder with OR gates and (ii) a 4-to-1 multiplexer.	10
Design a 4-bit binary adder-subtractor using full adders and XOR gates. Explain how the mode control line M selects between addition and subtraction, and how overflow is detected.	15
Explain the design procedure for combinational circuits. Using it, design a BCD-to-Excess-3 code converter, showing the truth table, K-maps for all four outputs (using don't-cares) and the final logic diagram.	15
Explain multiplexers. Show how a 2^n -to-1 MUX can realise any function of $n+1$ variables, and implement $F(A,B,C,D) = \Sigma m(0,1,3,4,8,9,15)$ using an 8-to-1 MUX.	15

MCQ traps — Combinational circuits

Trap	Truth
A 4-to-1 MUX has 4 select lines	It has 2 select lines ($2^n \rightarrow n$).
A decoder has n inputs and n outputs	n inputs, 2^n outputs.
An 8-to-1 MUX can implement 3-variable functions	It can implement 4-variable functions ($n+1$).
Half adder can be cascaded to add multi-bit numbers	It cannot — no carry input. That's the definition of "half".
Full subtractor difference $\neq$ full adder sum	They are identical : $A \oplus B \oplus C$. Only borrow/carry differ.
A ripple carry adder is fast	It is the slowest ; delay grows linearly with n. CLA fixes this.
Priority encoder and encoder are the same	Priority encoder resolves multiple simultaneous active inputs.
Decoder $\neq$ demultiplexer	A decoder with enable = demultiplexer .

Parity can correct a single-bit error	It can only detect an odd number of errors. Hamming corrects.
For n-bit binary→Gray you need n XOR gates	You need n-1 XORs (the MSB passes through unchanged).

PART E — SEQUENTIAL CIRCUITS